Linear Algebra & Matrix Methods for QED

A Comprehensive Review of Griffiths Chapter 7

With Supplementary Material from Thomson & Lancaster–Blundell

Prepared for self-study — all results derived from first principles

---

Contents

1 The Big Picture: Why Linear Algebra Runs QED **3**

2 The Pauli Matrices: 2×2 Building Blocks **3**
2.1 Definition and Explicit Form . . . 3
2.2 Essential Properties (All Provable by Direct Multiplication) . . . 3

3 The Dirac Gamma Matrices **4**
3.1 Definition: The Clifford Algebra . . . 4
3.2 Explicit Representation (Dirac–Pauli) . . . 4
3.3 Basic Properties from the Clifford Algebra . . . 5
3.4 The Metric Tensor $g^{\mu\nu}$ . . . 5

4 The Feynman Slash Notation **5**
4.1 Slash Product Identity . . . 5

5 Dirac Spinors as Vectors **6**
5.1 Structure of Spinors . . . 6
5.2 The Adjoint Spinor . . . 6
5.3 Bilinear Covariants: Scalars from Spinors . . . 6

6 Outer Products and Completeness Relations **7**
6.1 Inner Product vs. Outer Product . . . 7
6.2 The Completeness Relations . . . 7

7 Casimir's Trick: From Spin Sums to Traces **8**
7.1 Complex Conjugation of Spinor Sandwiches . . . 8
7.2 The Main Result . . . 8

8 General Properties of the Trace **9**

9 Contraction Theorems **10**
9.1 Proof of Rule 7: $\gamma_\mu\gamma^\nu\gamma^\mu = -2\gamma^\nu$ . . . 10
9.2 Slash Versions . . . 10
9.3 Proof of Rule 8′: $\gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$ . . . 11

10 Trace Theorems for Gamma Matrices **11**
10.1 Proofs of the Major Trace Theorems 11
10.1.1 Theorem 10: Trace of an Odd Number of Gamma Matrices = 0 12
10.1.2 Theorem 12: $\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$ 12
10.1.3 Theorem 13: $\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$ 12

11 The γ^5 Matrix and the Levi-Civita Symbol **12**
11.1 Definition and Properties of γ^5 13
11.2 The Levi-Civita Symbol $\varepsilon^{\mu\nu\rho\sigma}$ 13
11.3 Contraction Identities for ε 13

12 Photon Polarization Completeness **14**

13 Putting It All Together: A Worked Example **14**
13.1 Step 1: Write the Amplitude 14
13.2 Step 2: Square and Average 15
13.3 Step 3: Apply Casimir's Trick (Twice) 15
13.4 Step 4: Evaluate the Electron Trace 15
13.5 Step 5: Evaluate the Muon Trace 15
13.6 Step 6: Contract the Two Traces 15

14 Summary: The Linear Algebra Toolkit **16**

15 Practice Problems **17**

§1 The Big Picture: Why Linear Algebra Runs QED

Physical Motivation

In QED, every observable (cross section, decay rate) ultimately reduces to computing $|\mathcal{M}|^2$, the squared amplitude. The amplitude $\mathcal{M}$ is built from **spinors** (column/row vectors in a 4-dimensional space) and **gamma matrices** (4×4 matrices). Squaring, summing over spins, and simplifying *all* reduce to matrix algebra—specifically, to computing **traces** of products of gamma matrices.
The entire computational pipeline is:

$$\text{Feynman diagram} \longrightarrow \mathcal{M} \longrightarrow |\mathcal{M}|^2 \xrightarrow{\text{spin avg.}} \langle|\mathcal{M}|^2\rangle \xrightarrow{\text{Casimir's trick}} \text{Tr(gamma matrices)} \longrightarrow \text{number}$$

Every step in this chain is linear algebra.

§2 The Pauli Matrices: 2×2 Building Blocks

§2.1 Definition and Explicit Form

The Pauli spin matrices are the three Hermitian, unitary, traceless 2×2 matrices:

Pauli Matrices

$$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Together with the identity I_2, they form a complete basis for all 2×2 matrices.

§2.2 Essential Properties (All Provable by Direct Multiplication)

P1. Hermitian: $\sigma_i^\dagger = \sigma_i$ for $i = 1, 2, 3$.

P2. Unitary: $\sigma_i \sigma_i = I_2$, so $\sigma_i^{-1} = \sigma_i$.

P3. Traceless: $\text{Tr}(\sigma_i) = 0$ for each i.

P4. Determinant: $\det(\sigma_i) = -1$ for each i.

P5. Product rule:

Pauli Product Identity

$$\sigma_i \, \sigma_j = \delta_{ij} \, I_2 + i \, \varepsilon_{ijk} \, \sigma_k$$

This single identity encodes both the commutator and anticommutator:

$$[\sigma_i, \sigma_j] = 2i \, \varepsilon_{ijk} \, \sigma_k, \qquad \{\sigma_i, \sigma_j\} = 2\delta_{ij} \, I_2$$

P6. Dot-product identity: For any 3-vectors $\mathbf{a}$ and $\mathbf{b}$:

$$(\boldsymbol{\sigma}\cdot\mathbf{a})(\boldsymbol{\sigma}\cdot\mathbf{b}) = \mathbf{a}\cdot\mathbf{b}\, I_2 + i\,\boldsymbol{\sigma}\cdot(\mathbf{a}\times\mathbf{b})$$

Proof: Expand the left side: $\sigma_i a_i \sigma_j b_j = a_i b_j(\delta_{ij} I + i\varepsilon_{ijk}\sigma_k) = \mathbf{a}\cdot\mathbf{b}\, I + i\varepsilon_{ijk} a_i b_j \sigma_k$.

P7. Exponential: For a unit vector $\hat{n}$ and angle θ:

$$e^{i\theta\,\hat{n}\cdot\boldsymbol{\sigma}/2} = \cos\left(\frac{\theta}{2}\right) I_2 + i\,\hat{n}\cdot\boldsymbol{\sigma}\,\sin\left(\frac{\theta}{2}\right)$$

This is the $SU(2)$ rotation matrix for spin-$\frac{1}{2}$ particles.

Why Pauli Matrices Matter for Chapter 7

The Dirac γ-matrices are constructed *from* Pauli matrices. In the Dirac–Pauli representation:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

Every property of the γ-matrices traces back to Pauli matrix algebra.

§3 The Dirac Gamma Matrices

§3.1 Definition: The Clifford Algebra

The Defining Relation

The gamma matrices γ^μ ($\mu = 0,1,2,3$) are four 4×4 matrices satisfying the **Clifford algebra** (anticommutation relation):

$$\boxed{\{\gamma^\mu,\gamma^\nu\} \equiv \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}\, I_4}$$

where $g^{\mu\nu} = \mathrm{diag}(+1,-1,-1,-1)$ is the Minkowski metric.
This single relation determines *all* algebraic properties of the gamma matrices. We never need an explicit representation to derive trace theorems or contraction identities.

§3.2 Explicit Representation (Dirac–Pauli)

In the standard Dirac–Pauli representation used by Griffiths:

$$\gamma^0 = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \quad (i = 1,2,3)$$

Key Point

While an explicit representation is needed to write down spinors, **all** the trace theorems, contraction identities, and cross-section calculations can be done using only the Clifford algebra $\{\gamma^\mu,\gamma^\nu\} = 2g^{\mu\nu}$—no explicit matrices required. This is the power of abstract algebra.

§3.3 Basic Properties from the Clifford Algebra

Setting $\mu = \nu$ in the Clifford algebra:

$$(\gamma^0)^2 = g^{00} I = +I, \qquad (\gamma^i)^2 = g^{ii} I = -I \quad (i = 1, 2, 3)$$

Setting $\mu \neq \nu$: all distinct gamma matrices *anticommute*:

$$\gamma^\mu \gamma^\nu = -\gamma^\nu \gamma^\mu \quad (\mu \neq \nu)$$

Hermiticity Properties

$$(\gamma^0)^\dagger = \gamma^0, \qquad (\gamma^i)^\dagger = -\gamma^i$$

Compactly: $(\gamma^\mu)^\dagger = \gamma^0 \gamma^\mu \gamma^0$. This is essential for defining the adjoint spinor $\bar{u} = u^\dagger \gamma^0$.

§3.4 The Metric Tensor $g^{\mu\nu}$

Metric Tensor Properties

The Minkowski metric $g^{\mu\nu} = g_{\mu\nu} = \text{diag}(+1, -1, -1, -1)$ satisfies:

$$g_{\mu\nu}\, g^{\nu\rho} = \delta_\mu^{\ \rho} \quad \text{(identity when contracted)} \tag{1}$$

$$g^{\mu\nu}\, g_{\mu\nu} = g_\mu^{\ \mu} = \delta_\mu^{\ \mu} = 4 \quad \text{(trace of identity in 4D)} \tag{2}$$

The factor of **4** appears everywhere in trace calculations—it's just the dimensionality of spacetime.

§4 The Feynman Slash Notation

Definition of Slash

For any four-vector a^μ, the "Feynman slash" is the 4×4 matrix:

$$\not{a} \equiv \gamma^\mu a_\mu = \gamma^0 a_0 - \gamma^1 a_1 - \gamma^2 a_2 - \gamma^3 a_3$$

Equivalently, $\not{a} = \gamma^\mu a_\mu = \gamma_\mu a^\mu$.

The slash notation converts a four-vector (a set of 4 numbers) into a 4×4 matrix. This is the bridge between kinematics (four-momenta) and the matrix algebra of QED.

§4.1 Slash Product Identity

From the Clifford algebra:

$$\not{a}\not{b} + \not{b}\not{a} = \gamma^\mu \gamma^\nu a_\mu b_\nu + \gamma^\nu \gamma^\mu a_\mu b_\nu = \{\gamma^\mu, \gamma^\nu\}\, a_\mu b_\nu = 2 g^{\mu\nu} a_\mu b_\nu = 2(a \cdot b)\, I$$

Slash Anticommutator — Griffiths Rule 5′

$$\boxed{\not{a}\not{b} + \not{b}\not{a} = 2(a \cdot b)\, I}$$

As a special case: $\not{a}\not{a} = a^2\, I = a^\mu a_\mu\, I$.
For a massive particle on shell: $\not{p}\not{p} = p^2 I = m^2 c^2\, I$.

§5 Dirac Spinors as Vectors

§5.1 Structure of Spinors

Dirac spinors are **four-component column vectors** in "spinor space." For a particle with four-momentum p^μ and spin state s:

$$u^{(s)}(p) = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{pmatrix} \quad \text{(particle)}, \qquad v^{(s)}(p) = \begin{pmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{pmatrix} \quad \text{(antiparticle)}$$

In the Dirac–Pauli representation, for a particle moving along $\hat{z}$ with energy E and momentum p:

$$u^{(1)} = \sqrt{E + mc^2} \begin{pmatrix} 1 \\ 0 \\ \frac{pc}{E+mc^2} \\ 0 \end{pmatrix}, \qquad u^{(2)} = \sqrt{E + mc^2} \begin{pmatrix} 0 \\ 1 \\ 0 \\ \frac{-pc}{E+mc^2} \end{pmatrix}$$

§5.2 The Adjoint Spinor

Adjoint Spinor

The **adjoint** (Dirac adjoint) of a spinor u is the 1×4 row vector:

$$\bar{u} \equiv u^\dagger \gamma^0$$

This is *not* just the Hermitian conjugate $u^\dagger$! The factor of γ^0 is required to make bilinears like $\bar{u}\,\gamma^\mu\, u$ transform properly as four-vectors under Lorentz transformations.

§5.3 Bilinear Covariants: Scalars from Spinors

The combination $\bar{u}(a)\,\Gamma\,u(b)$ (where Γ is some 4×4 matrix) is a 1×1 matrix, i.e., a **complex number**:

$$\underbrace{\bar{u}(a)}_{1\times4}\ \underbrace{\Gamma}_{4\times4}\ \underbrace{u(b)}_{4\times1} = \text{complex number}$$

This is why we can rearrange spinor components freely when doing spin sums—they are just numbers, and numbers commute.

The Rearrangement Trick (Griffiths' "fancy footwork")

Since $\bar{u}_i \, \Gamma_{ij} \, u_j$ is just a product of numbers (with indices tracking matrix multiplication), we can freely rearrange:

$$\bar{u}(a)_i \cdot u(b)_j = u(b)_j \cdot \bar{u}(a)_i$$

But if we view $u(b)_j \, \bar{u}(a)_i$ as the (j, i) element of the **outer product** matrix $u(b) \, \bar{u}(a)$, this is the key step that converts spin sums into traces.

§6 Outer Products and Completeness Relations

§6.1 Inner Product vs. Outer Product

This distinction is the conceptual core of the trace method:

Two Ways to Multiply a Column by a Row

Inner product: Row $\times$ Column $= 1 \times 1$ scalar:

$$\bar{u} \, u = \sum_{i=1}^{4} \bar{u}_i \, u_i \; \in \; \mathbb{C}$$

Outer product: Column $\times$ Row $= 4 \times 4$ matrix:

$$u \, \bar{u} = \begin{pmatrix} u_1\bar{u}_1 & u_1\bar{u}_2 & u_1\bar{u}_3 & u_1\bar{u}_4 \\ u_2\bar{u}_1 & \cdots & & \\ u_3\bar{u}_1 & & \cdots & \\ u_4\bar{u}_1 & & & u_4\bar{u}_4 \end{pmatrix}$$

The (i, j) element of $u \, \bar{u}$ is $u_i \, \bar{u}_j$.

§6.2 The Completeness Relations

When we sum the outer product over both spin states, we get a remarkably simple result:

Completeness Relations — Griffiths Eq. 7.99

$$\boxed{\sum_{s=1}^{2} u^{(s)}(p) \, \bar{u}^{(s)}(p) = \not{p} + mc \, I}$$

$$\boxed{\sum_{s=1}^{2} v^{(s)}(p) \, \bar{v}^{(s)}(p) = \not{p} - mc \, I}$$

Here $\not{p} + mc$ and $\not{p} - mc$ are 4×4 matrices. The sign difference between the u and v completeness relations is physically significant (particles vs. antiparticles).

Physical Meaning of Completeness

The completeness relation says: "If I don't know the spin of the particle, then summing over all spin possibilities gives me a projection operator proportional to $\not{p} + m$." It's the spinor analog of $\sum_m |l,m\rangle\langle l,m| = \mathbf{1}$ in quantum mechanics—resolution of the identity in spin space.

Proof Sketch (from Griffiths Problem 7.24)

Using the explicit spinors $u^{(1)}, u^{(2)}$ and the Dirac–Pauli representation:

$$\sum_{s=1}^{2} u^{(s)}\bar{u}^{(s)} = (E+mc^2)\sum_{s=1}^{2}\begin{pmatrix}\phi_s \\ \frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+mc^2}\phi_s\end{pmatrix}\begin{pmatrix}\phi_s^\dagger & -\phi_s^\dagger\frac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+mc^2}\end{pmatrix}$$

Using $\sum_s \phi_s\phi_s^\dagger = I_2$ (the 2×2 identity, since $\phi_1 = \binom{1}{0}$, $\phi_2 = \binom{0}{1}$) and $(\boldsymbol{\sigma}\cdot\mathbf{p})^2 = |\mathbf{p}|^2 = E^2 - m^2c^4$:

$$= \begin{pmatrix}(E+mc^2)I_2 & -\boldsymbol{\sigma}\cdot\mathbf{p} \\ \boldsymbol{\sigma}\cdot\mathbf{p} & -(E-mc^2)I_2\end{pmatrix} = \gamma^0 E - \gamma^i p_i + mc^2\, I = \not{p} + mc$$

where we used $\not{p} = \gamma^\mu p_\mu = \gamma^0 E - \boldsymbol{\gamma}\cdot\mathbf{p}$ and the explicit forms of γ^0, γ^i.

§7 Casimir's Trick: From Spin Sums to Traces

What We Want

Given a squared amplitude like $|\mathcal{M}|^2 \sim [\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma^\nu u(2)] \times \text{c.c.}$, we want to sum over all initial and final spins to get $\langle|\mathcal{M}|^2\rangle$ **without** computing individual spinor products.

§7.1 Complex Conjugation of Spinor Sandwiches

The first step: what is $[\bar{u}(a)\,\Gamma\,u(b)]^*$?

The "Bar" of a Matrix

For any 4×4 matrix Γ (built from gamma matrices), define:

$$\bar{\Gamma} \equiv \gamma^0\,\Gamma^\dagger\,\gamma^0$$

Then:

$$[\bar{u}(a)\,\Gamma\,u(b)]^* = \bar{u}(b)\,\bar{\Gamma}\,u(a)$$

Proof: $[\bar{u}\Gamma u]^* = [\bar{u}\Gamma u]^\dagger = u^\dagger\Gamma^\dagger(\bar{u})^\dagger = u^\dagger\Gamma^\dagger\gamma^{0\dagger}u = u^\dagger\gamma^0\gamma^0\Gamma^\dagger\gamma^0 u = \bar{u}\,\bar{\Gamma}\,u$.
Key example: For $\Gamma = \gamma^\mu$, we have $\bar{\Gamma} = \gamma^0(\gamma^\mu)^\dagger\gamma^0 = \gamma^\mu$. So $\overline{\gamma^\mu} = \gamma^\mu$.
In fact, $\bar{\Gamma} = \Gamma$ for *all* Standard Model vertex factors.

§7.2 The Main Result

Consider the generic quantity:

$$G \equiv [\bar{u}(a)\,\Gamma_1\,u(b)]\;[\bar{u}(b)\,\Gamma_2\,u(a)]$$

(This arises when you multiply $\mathcal{M}$ by $\mathcal{M}^*$ and use $\bar{\Gamma} = \Gamma$.)

Casimir's Trick — Griffiths Eq. 7.125

$$\sum_{\text{all spins}} [\bar{u}(a)\,\Gamma_1\,u(b)]\,[\bar{u}(a)\,\Gamma_2\,u(b)]^* = \text{Tr}\Big[\Gamma_1(\not{p}_b + m_b c)\,\bar{\Gamma}_2\,(\not{p}_a + m_a c)\Big]$$

If either u is replaced by a v spinor, the corresponding mass term flips sign (from $+m$ to $-m$).

Derivation (Step by Step)

Step 1: Write $G = [\bar{u}(a)\,\Gamma_1\,u(b)][\bar{u}(b)\,\Gamma_2\,u(a)]$ with explicit indices:

$$G = \bar{u}(a)_i\,(\Gamma_1)_{ij}\,u(b)_j\,\bar{u}(b)_n\,(\Gamma_2)_{nm}\,u(a)_m$$

All of these are just numbers, so we can reorder them freely.

Step 2: Sum over spins of particle b:

$$\sum_{s_b} u(b)_j\,\bar{u}(b)_n = (\not{p}_b + m_b c)_{jn}$$

This is the completeness relation used as a 4×4 matrix element.

Step 3: The expression becomes:

$$\sum_{s_b} G = \bar{u}(a)_i\,(\Gamma_1)_{ij}\,(\not{p}_b + m_b)_{jn}\,(\Gamma_2)_{nm}\,u(a)_m \equiv \bar{u}(a)_i\,Q_{im}\,u(a)_m$$

where $Q = \Gamma_1(\not{p}_b + m_b)\Gamma_2$ is a 4×4 matrix.

Step 4: Now sum over spins of particle a:

$$\sum_{s_a} \bar{u}(a)_i\,Q_{im}\,u(a)_m = \sum_{s_a} u(a)_m\,\bar{u}(a)_i\,Q_{im} = (\not{p}_a + m_a)_{mi}\,Q_{im} = [Q(\not{p}_a + m_a)]_{ii}$$

The last step is just matrix multiplication summed over diagonal elements = Tr.

Step 5: Therefore:

$$\sum_{\text{spins}} G = \text{Tr}\Big[\Gamma_1(\not{p}_b + m_b)\,\Gamma_2\,(\not{p}_a + m_a)\Big] \qquad \square$$

The Magic of Casimir's Trick

- **No spinors survive.** After the spin sum, everything is a trace of 4×4 matrices.
- **No representation needed.** The trace can be evaluated purely from the Clifford algebra.
- The outer product $u\bar{u} = \not{p} + m$ is the linchpin: it converts a sum of number×number products into a single matrix equation.

§8 General Properties of the Trace

Before diving into gamma-matrix trace theorems, recall the general properties of the trace of *any* square matrix:

General Trace Properties

For any $n \times n$ matrices A, B and scalar a:

1. Linearity: $$\mathrm{Tr}(A+B) = \mathrm{Tr}(A) + \mathrm{Tr}(B) \tag{3}$$

2. Scalar pull-out: $$\mathrm{Tr}(aA) = a\ \mathrm{Tr}(A) \tag{4}$$

3. Cyclic property: $$\mathrm{Tr}(AB) = \mathrm{Tr}(BA) \tag{5}$$

$$\mathrm{Tr}(ABC) = \mathrm{Tr}(CAB) = \mathrm{Tr}(BCA) \tag{6}$$

$$\neq \mathrm{Tr}(ACB) \text{ in general!} \tag{7}$$

Cyclic $\neq$ Arbitrary Reordering

You can "peel" matrices from the *back* and move them to the *front*, preserving order. Think of the matrices as beads on a necklace—you can rotate the necklace, but not rearrange the beads.

$$\mathrm{Tr}(ABCD) = \mathrm{Tr}(DABC) = \mathrm{Tr}(CDAB) = \mathrm{Tr}(BCDA) \neq \mathrm{Tr}(ABDC)$$

§9 Contraction Theorems

The **contraction theorems** tell you what happens when you "contract" a gamma matrix with itself using $\gamma_\mu \ldots \gamma^\mu$ (i.e., sum over μ). These are proved by repeated application of the Clifford algebra.

Contraction Theorems — Griffiths Rules 6–9

6. $$\gamma_\mu \gamma^\mu = g_{\mu\nu}\gamma^\mu\gamma^\nu = g^\mu_{\ \mu} I = 4I \tag{8}$$

7. $$\gamma_\mu \gamma^\nu \gamma^\mu = -2\gamma^\nu \tag{9}$$

8. $$\gamma_\mu \gamma^\nu\gamma^\lambda \gamma^\mu = 4\, g^{\nu\lambda} I \tag{10}$$

9. $$\gamma_\mu \gamma^\nu\gamma^\lambda\gamma^\sigma \gamma^\mu = -2\,\gamma^\sigma\gamma^\lambda\gamma^\nu \tag{11}$$

§9.1 Proof of Rule 7: $\gamma_\mu\gamma^\nu\gamma^\mu = -2\gamma^\nu$

Start from the Clifford algebra:

$$\begin{aligned} \gamma_\mu \gamma^\nu \gamma^\mu &= \gamma_\mu(2g^{\mu\nu} - \gamma^\mu\gamma^\nu) \\ &= 2g^{\mu\nu}\gamma_\mu - \gamma_\mu\gamma^\mu\gamma^\nu \\ &= 2\gamma^\nu - 4\gamma^\nu \quad \text{(using Rule 6)} \\ &= -2\gamma^\nu \qquad \square \end{aligned} \tag{12}$$

§9.2 Slash Versions

The same identities hold with slashed quantities:

Slash Contraction Theorems — Griffiths Rules 7′–9′

$$\textbf{7}'. \qquad \gamma_\mu \not{a} \gamma^\mu = -2\not{a} \tag{13}$$

$$\textbf{8}'. \qquad \gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)\, I \tag{14}$$

$$\textbf{9}'. \qquad \gamma_\mu \not{a} \not{b} \not{c} \gamma^\mu = -2\not{c} \not{b} \not{a} \tag{15}$$

These follow by substituting $\gamma^\nu = \gamma^\nu a_\nu / a_\nu$ (schematically) into Rules 7–9, or more carefully, by writing $\not{a} = \gamma^\nu a_\nu$ and using the unslashed rules.

§9.3 Proof of Rule 8′: $\gamma_\mu \not{a} \not{b} \gamma^\mu = 4(a \cdot b)$

$$\begin{aligned} \gamma_\mu \not{a} \not{b} \gamma^\mu &= \gamma_\mu \gamma^\nu a_\nu \gamma^\lambda b_\lambda \gamma^\mu = a_\nu b_\lambda \, \gamma_\mu \gamma^\nu \gamma^\lambda \gamma^\mu \\ &= a_\nu b_\lambda \, (4 g^{\nu\lambda}) \quad \text{(Rule 8)} \\ &= 4\, a_\nu b^\nu = 4(a \cdot b) \qquad \square \end{aligned} \tag{16}$$

§10 Trace Theorems for Gamma Matrices

These are the workhorses of QED calculations. All are proved from the Clifford algebra alone.

Complete Set of Trace Theorems — Griffiths Rules 10–16

10. $\mathrm{Tr}(\text{odd \# of } \gamma\text{'s}) = 0$ (17)

11. $\mathrm{Tr}(I) = 4$ (18)

12. $\mathrm{Tr}(\gamma^\mu \gamma^\nu) = 4\, g^{\mu\nu}$ (19)

12′. $\mathrm{Tr}(\not{a}\not{b}) = 4(a \cdot b)$ (20)

13. $\mathrm{Tr}(\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4(g^{\mu\nu} g^{\rho\sigma} - g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho})$ (21)

13′. $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d}) = 4\big[(a \cdot b)(c \cdot d) - (a \cdot c)(b \cdot d) + (a \cdot d)(b \cdot c)\big]$ (22)

14. $\mathrm{Tr}(\gamma^5) = 0$ (23)

15. $\mathrm{Tr}(\gamma^5 \gamma^\mu \gamma^\nu) = 0$ (24)

16. $\mathrm{Tr}(\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma) = 4i\, \varepsilon^{\mu\nu\rho\sigma}$ (25)

§10.1 Proofs of the Major Trace Theorems

§10.1.1 Theorem 10: Trace of an Odd Number of Gamma Matrices $= 0$

Proof Using γ^5

The proof uses $\gamma^5\gamma^5 = I$ and the fact that γ^5 anticommutes with every γ^μ:

$$\gamma^5\gamma^\mu = -\gamma^\mu\gamma^5$$

Consider the trace of n gamma matrices (with n odd):

$$\begin{aligned}
\operatorname{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}) &= \operatorname{Tr}(\gamma^5\gamma^5\,\gamma^{\mu_1}\cdots\gamma^{\mu_n}) \quad \text{(insert } I = \gamma^5\gamma^5\text{)} \\
&= \operatorname{Tr}(\gamma^5\,\gamma^{\mu_1}\cdots\gamma^{\mu_n}\,\gamma^5) \quad \text{(cyclic property)} \\
&= (-1)^n\ \operatorname{Tr}(\gamma^5\gamma^5\,\gamma^{\mu_1}\cdots\gamma^{\mu_n}) \quad \text{(anticommute } \gamma^5 \text{ through } n\ \gamma\text{'s)} \\
&= (-1)^n\ \operatorname{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}) \qquad (26)
\end{aligned}$$

For n odd, $(-1)^n = -1$, so $\operatorname{Tr} = -\operatorname{Tr}$, which implies $\operatorname{Tr} = 0$. □

§10.1.2 Theorem 12: $\operatorname{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$

From the Clifford algebra: $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}I$. Take the trace:

$$\operatorname{Tr}(\gamma^\mu\gamma^\nu) + \operatorname{Tr}(\gamma^\nu\gamma^\mu) = 2g^{\mu\nu}\operatorname{Tr}(I) = 8g^{\mu\nu}$$

By the cyclic property, $\operatorname{Tr}(\gamma^\mu\gamma^\nu) = \operatorname{Tr}(\gamma^\nu\gamma^\mu)$, so $2\operatorname{Tr}(\gamma^\mu\gamma^\nu) = 8g^{\mu\nu}$, giving $\operatorname{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}$. □

§10.1.3 Theorem 13: $\operatorname{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$

Proof by Repeated Anticommutation

Use $\gamma^\mu\gamma^\nu = 2g^{\mu\nu} - \gamma^\nu\gamma^\mu$ to "walk" the first γ through to the end:

$$\begin{aligned}
\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma &= (2g^{\mu\nu} - \gamma^\nu\gamma^\mu)\gamma^\rho\gamma^\sigma \\
&= 2g^{\mu\nu}\gamma^\rho\gamma^\sigma - \gamma^\nu(2g^{\mu\rho} - \gamma^\rho\gamma^\mu)\gamma^\sigma \\
&= 2g^{\mu\nu}\gamma^\rho\gamma^\sigma - 2g^{\mu\rho}\gamma^\nu\gamma^\sigma + \gamma^\nu\gamma^\rho(2g^{\mu\sigma} - \gamma^\sigma\gamma^\mu) \\
&= 2g^{\mu\nu}\gamma^\rho\gamma^\sigma - 2g^{\mu\rho}\gamma^\nu\gamma^\sigma + 2g^{\mu\sigma}\gamma^\nu\gamma^\rho - \gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu \qquad (27)
\end{aligned}$$

Now take the trace, using cyclic property on the last term:

$$\operatorname{Tr}(\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu) = \operatorname{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) \quad \text{(cycling } \gamma^\mu \text{ from back to front)}$$

So:

$$2\operatorname{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 2g^{\mu\nu}\underbrace{\operatorname{Tr}(\gamma^\rho\gamma^\sigma)}_{4g^{\rho\sigma}} - 2g^{\mu\rho}\underbrace{\operatorname{Tr}(\gamma^\nu\gamma^\sigma)}_{4g^{\nu\sigma}} + 2g^{\mu\sigma}\underbrace{\operatorname{Tr}(\gamma^\nu\gamma^\rho)}_{4g^{\nu\rho}}$$

Dividing by 2 gives the result. □

§11 The γ^5 Matrix and the Levi-Civita Symbol

§11.1 Definition and Properties of γ^5

Definition of γ^5

$$\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$$

In the Dirac–Pauli representation:

$$\gamma^5 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

Key properties (all follow from the Clifford algebra):

(a) $(\gamma^5)^2 = I$ (it squares to the identity)

(b) $(\gamma^5)^\dagger = \gamma^5$ (Hermitian)

(c) $\{\gamma^5, \gamma^\mu\} = 0$ for all μ (γ^5 **anticommutes** with every γ^μ)

(d) $\mathrm{Tr}(\gamma^5) = 0$

Property (c) is the crucial one. *Proof:* γ^5 is the product of all four distinct γ-matrices (times i). To commute γ^μ past γ^5, we must anticommute it past three *other* γ-matrices (since γ^μ commutes with itself trivially), picking up $(-1)^3 = -1$.

§11.2 The Levi-Civita Symbol $\varepsilon^{\mu\nu\rho\sigma}$

Definition

The totally antisymmetric Levi-Civita symbol in 4 dimensions:

$$\varepsilon^{\mu\nu\rho\sigma} = \begin{cases} +1 & \text{if } (\mu\nu\rho\sigma) \text{ is an } \textit{even} \text{ permutation of } (0123) \\ -1 & \text{if } (\mu\nu\rho\sigma) \text{ is an } \textit{odd} \text{ permutation of } (0123) \\ 0 & \text{if any two indices are equal} \end{cases}$$

Warning (Griffiths' convention): $\varepsilon^{0123} = +1$, but $\varepsilon_{0123} = -1$ (three spatial indices are lowered, each introducing a minus sign from $g_{ii} = -1$). Different textbooks use different sign conventions!

§11.3 Contraction Identities for ε

Levi-Civita Contractions — Griffiths Problem 7.35

$$\varepsilon^{\alpha\beta\gamma\delta}\,\varepsilon_{\alpha\mu\nu\lambda} = -\det\begin{pmatrix} \delta^\beta_\mu & \delta^\beta_\nu & \delta^\beta_\lambda \\ \delta^\gamma_\mu & \delta^\gamma_\nu & \delta^\gamma_\lambda \\ \delta^\delta_\mu & \delta^\delta_\nu & \delta^\delta_\lambda \end{pmatrix} \tag{28}$$

$$\varepsilon^{\alpha\beta\gamma\delta}\,\varepsilon_{\alpha\beta\nu\lambda} = -2(\delta^\gamma_\nu\delta^\delta_\lambda - \delta^\gamma_\lambda\delta^\delta_\nu) \tag{29}$$

$$\varepsilon^{\alpha\beta\gamma\delta}\,\varepsilon_{\alpha\beta\gamma\lambda} = -6\,\delta^\delta_\lambda \tag{30}$$

$$\varepsilon^{\alpha\beta\gamma\delta}\,\varepsilon_{\alpha\beta\gamma\delta} = -24 \tag{31}$$

The minus signs come from the Minkowski-metric lowering of indices.

These identities are needed whenever γ^5 appears in a trace (e.g., in weak interaction calculations).

§12 Photon Polarization Completeness

Just as fermion spin sums use the spinor completeness relation, photon polarization sums use:

Photon Polarization Sum — Griffiths Eq. 7.105

$$\boxed{\sum_{\lambda=1}^{2} \epsilon^{\mu}_{(\lambda)}\,\epsilon^{*\nu}_{(\lambda)} \;\longrightarrow\; -g^{\mu\nu}}$$

Why the Arrow, Not Equals?

This replacement is valid *only* inside a gauge-invariant amplitude. The full completeness relation includes terms involving the photon momentum k^μ, but gauge invariance (specifically, $k_\mu\mathcal{M}^\mu = 0$) guarantees these extra terms vanish. So for QED calculations, the simple replacement $\to -g^{\mu\nu}$ suffices.

§13 Putting It All Together: A Worked Example

Example: Electron–Muon Scattering $e^-\mu^- \to e^-\mu^-$

Compute the spin-averaged matrix element squared $\langle|\mathcal{M}|^2\rangle$ for electron–muon scattering.

§13.1 Step 1: Write the Amplitude

From the Feynman rules:

$$\mathcal{M} = \frac{-g_e^2}{(p_1 - p_3)^2}\,[\bar{u}(3)\gamma^\mu u(1)]\,[\bar{u}(4)\gamma_\mu u(2)]$$

§13.2 Step 2: Square and Average

$$\langle|\mathcal{M}|^2\rangle = \frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{g_e^4}{4(p_1-p_3)^4}\sum_{\text{spins}}[\bar{u}(3)\gamma^\mu u(1)][\bar{u}(4)\gamma_\mu u(2)][\bar{u}(3)\gamma^\nu u(1)]^*[\bar{u}(4)\gamma_\nu u(2)]^*$$

The factor of $\frac{1}{4}$ averages over the 2 initial electron spins $\times$ 2 initial muon spins.

§13.3 Step 3: Apply Casimir's Trick (Twice)

The spin sum factors into two independent traces:

$$\sum_{\text{spins}} = \underbrace{\text{Tr}[\gamma^\mu(\not{p}_1 + mc)\gamma^\nu(\not{p}_3 + mc)]}_{\text{electron trace}} \times \underbrace{\text{Tr}[\gamma_\mu(\not{p}_2 + Mc)\gamma_\nu(\not{p}_4 + Mc)]}_{\text{muon trace}}$$

§13.4 Step 4: Evaluate the Electron Trace

Expand:

$$\text{Tr}[\gamma^\mu(\not{p}_1 + m)\gamma^\nu(\not{p}_3 + m)] = \text{Tr}(\gamma^\mu\not{p}_1\gamma^\nu\not{p}_3) + m^2\ \text{Tr}(\gamma^\mu\gamma^\nu)$$

(The cross terms $\text{Tr}(\gamma^\mu\not{p}_1\gamma^\nu m)$ and $\text{Tr}(\gamma^\mu m\gamma^\nu\not{p}_3)$ vanish by Rule 10—odd number of γ's.)

Using Rule 13′ on the first term and Rule 12 on the second:

$$\text{Tr}(\gamma^\mu\not{p}_1\gamma^\nu\not{p}_3) = 4[p_1^\mu p_3^\nu - g^{\mu\nu}(p_1\cdot p_3) + p_1^\nu p_3^\mu] \tag{32}$$

$$m^2\ \text{Tr}(\gamma^\mu\gamma^\nu) = 4m^2 g^{\mu\nu} \tag{33}$$

Electron Trace Result

$$\text{Tr}[\gamma^\mu(\not{p}_1 + m)\gamma^\nu(\not{p}_3 + m)] = 4\big[p_1^\mu p_3^\nu + p_1^\nu p_3^\mu + g^{\mu\nu}(m^2 - p_1\cdot p_3)\big]$$

§13.5 Step 5: Evaluate the Muon Trace

By the same logic (with $m \to M$, $1 \to 2$, $3 \to 4$, and μ, ν lowered):

$$\text{Tr}[\gamma_\mu(\not{p}_2 + M)\gamma_\nu(\not{p}_4 + M)] = 4\big[p_{2\mu}\,p_{4\nu} + p_{2\nu}\,p_{4\mu} + g_{\mu\nu}(M^2 - p_2\cdot p_4)\big]$$

§13.6 Step 6: Contract the Two Traces

Multiply the electron and muon traces and contract μ, ν indices. This requires:

$$p_1^\mu p_3^\nu\ p_{2\mu}p_{4\nu} = (p_1\cdot p_2)(p_3\cdot p_4) \tag{34}$$

$$g^{\mu\nu}\,g_{\mu\nu} = 4 \tag{35}$$

$$p_1^\mu\,g_{\mu\nu} = p_{1\nu} \quad \text{etc.} \tag{36}$$

After careful contraction of all terms (12 terms from the electron trace $\times$ the muon trace):

Final Result — Griffiths Eq. 7.129

$$\langle|\mathcal{M}|^2\rangle = \frac{g_e^4}{(p_1-p_3)^4}\left[(p_1\cdot p_2)(p_3\cdot p_4)+(p_1\cdot p_4)(p_2\cdot p_3)-m^2(p_2\cdot p_4)-M^2(p_1\cdot p_3)+2m^2M^2\right]$$

Observe the Structure

The result is a **Lorentz scalar** built entirely from dot products of four-momenta. No spinors, no gamma matrices, no representation-dependent quantities survive. The entire machinery of linear algebra has been deployed to convert a quantum-mechanical amplitude into a kinematic formula that can be evaluated in any reference frame.

§14 Summary: The Linear Algebra Toolkit

Tool	What It Is	Where Used
Pauli matrices σ_i	2×2 Hermitian matrices	Build γ-matrices
Gamma matrices γ^μ	4×4 matrices, Clifford algebra	Vertices, propagators
Slash notation $\not{a}$	$\gamma^\mu a_\mu$: 4-vector $\to$ matrix	Propagators, completeness
Spinors u, v	4-component column vectors	External fermion lines
Adjoint $\bar{u} = u^\dagger\gamma^0$	4-component row vector	Bilinear covariants
Outer product $u\bar{u}$	4×4 matrix from spinors	Completeness relations
Completeness $\sum u\bar{u} = \not{p} + m$	Spin sum $\to$ matrix	Casimir's trick
Casimir's trick	Spin sum $\to$ trace	$\langle\|\mathcal{M}\|^2\rangle$ calculation
Trace theorems	Evaluate $\mathrm{Tr}(\gamma\cdots\gamma)$	Final simplification
Contraction theorems	$\gamma_\mu\cdots\gamma^\mu$	Intermediate simplification
γ^5 and $\varepsilon^{\mu\nu\rho\sigma}$	Chirality, parity violation	Weak interactions
Polarization sum $\sum\epsilon\epsilon^* \to -g$	Photon completeness	Photon spin averaging

The Conceptual Flow

Every QED calculation follows the same linear algebra pipeline:
1. Feynman rules give $\mathcal{M}$ as products of spinors and γ-matrices (matrix algebra).
2. $|\mathcal{M}|^2$ involves complex conjugation $\to$ use $\bar{\Gamma} = \gamma^0\Gamma^\dagger\gamma^0$ (Hermitian conjugation of matrices).
3. Spin averaging uses completeness relations $\sum u\bar{u} = \not{p} + m$ (outer products $\to$ projection operators).
4. Casimir's trick converts spin sums to traces (the trace replaces explicit spinor compo-

nents).
5. Trace theorems evaluate $\mathrm{Tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n})$ (abstract algebra, no representation needed).
6. Index contraction with $g^{\mu\nu}$ yields Lorentz-invariant dot products (tensor algebra).
7. Kinematics converts dot products to masses, energies, angles (physics!).

§15 Practice Problems

Problem 1: Contraction Warm-up

Prove that $\gamma_\mu\gamma^\nu\gamma^\lambda\gamma^\mu = 4g^{\nu\lambda}$ (Rule 8) by using only the Clifford algebra and Rule 7.

Problem 2: Four-Gamma Trace

Starting from the Clifford algebra, prove Trace Theorem 13:

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(g^{\mu\nu}g^{\rho\sigma} - g^{\mu\rho}g^{\nu\sigma} + g^{\mu\sigma}g^{\nu\rho})$$

without looking at the proof in Section 9.3.

Problem 3: Slash Trace

Evaluate $\mathrm{Tr}(\not{a}\not{b}\not{c}\not{d})$ where a, b, c, d are arbitrary four-vectors. Verify that the result is symmetric under simultaneous interchange of (a, b) and (c, d).

Problem 4: Completeness Check

Using the explicit spinors for a particle at rest ($\mathbf{p} = 0$, mass m), verify that $\sum_s u^{(s)}u^{(s)\dagger}\gamma^0 = \gamma^0 m + m\,I$, which is $\not{p} + m$ at rest.

Problem 5: The Full Electron–Muon Scattering

Fill in the 12-term contraction in Section 11, Step 6. Show explicitly which terms cancel and which combine to give the final result (Eq. 7.129).

Problem 6: γ^5 Trace

Prove that $\mathrm{Tr}(\gamma^5\gamma^\mu\gamma^\nu) = 0$ (Theorem 15). *Hint:* Use the same γ^5-insertion trick as in the proof of Theorem 10.

Problem 7: Levi-Civita Contraction

Using the definition of $\varepsilon^{\mu\nu\rho\sigma}$, prove that $\varepsilon^{\alpha\beta\gamma\delta}\varepsilon_{\alpha\beta\gamma\delta} = -24$.

Document prepared for self-study in Elementary Particle Physics.
Primary source: Griffiths, *Introduction to Elementary Particles*, Chapter 7.
Supplementary: Thomson, *Modern Particle Physics*, Chapter 6;
Lancaster & Blundell, *QFT for the Gifted Amateur*, Chapters 37 & 40.